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Deep learning based multi-temporal crop classification

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Highlights

- Deep neural networks were developed for crop classification.
- Deep neural network achieved 85.54% accuracy and an F1 score of 0.73.
- The best non-deep-learning classifier achieved 84.17% accuracy and an F1 score of 0.69.
- One-dimensional convolutional neural network was used as automated temporal feature extractor.
- One-dimensional convolutional neural network identifies complex seasonal dynamics of economic crops.

Abstract

This study aims to develop a deep learning based classification framework for remotely sensed time series. The experiment was carried out in Yolo County, California, which has a very diverse irrigated agricultural system dominated by economic crops. For the challenging task of classifying summer crops using Landsat Enhanced Vegetation Index (EVI) time series, two types of deep learning models were designed: one is based on Long Short-Term Memory (LSTM), and the other is based on one-dimensional convolutional (Conv1D) layers. Three widely-used classifiers were also tested for comparison, including a gradient boosting machine called XGBoost, Random Forest, and Support Vector Machine. Although LSTM is widely used for sequential data representation, in this study its accuracy (82.41%) and F1 score (0.67) were the lowest among all the mainly consists of a stack of Conv1D layers and an inception module. The behavior of the Conv1D-based model was inspected by visualizing the activation on different layers. The model employs EVI time series by examining shapes at various scales in a hierarchical manner. Lower Conv1D layers of the optimized model capture small scale temporal variations, while upper layers focus on overall seasonal patterns. Conv1D layers were used as an embedded multi-level feature extractor in the classification model which automatically extracts features from input time series during training. The automated feature extraction reduces the dependency on manual feature engineering and pre-defined equations of crop growing cycles. This study shows that the Conv1D-based deep learning framework provides an effective and efficient method of time series representation in multi-temporal classification tasks.

Keywords

Crop classification; Deep learning; Artificial neural network; Convolutional neural network; Multi-temporal classification; Landsat

Previous article in this issue Next article in this issue

Introduction

Seasonality is one of the most prominent characteristics of vegetation. Multi-temporal remote sensing is an efficient source of time series observations to monitor growing dynamics for vegetation classification (Rogan et al., 2002; Xie et al., 2008). As remotely-sensed time series are being generated at an unprecedented scale and rate from an expanding collection of platforms, a key aspect towards the goal of classification is how to use time series to fully utilize the wealth of seasonal patterns and sequential relationships. As a consequence, there has been an increasing amount of interest in time series representation, which extracts features from time series to retrieve useful information on vegetation growing stages and conditions.

In multi-temporal remote sensing, various methods have been developed to process and analyze vegetation index (VI) time series. Some studies directly used original VI values during different periods of the year as the main input to rule-based classification algorithms like decision trees (Lloyd, 1990; Friedl et al., 1999; Wardlow and Egbert, 2008). The direct use of time series is simple but effective for vegetation types with distinctive temporal characteristics like rice (Xiao et al., 2005, Xiao et al., 2006, Dong et al., 2015, G. Zhang et al., 2015). In these methods, the sequential relationship of multi-temporal observations was not explicitly considered, which might ignore some useful information in time series inputs.

values to calculate metrics such as the maximum VI, time of peak VI, and onset of green-up to characterize VI time series (Reed et al., 1994; Vina et al., 2004; Brown et al., 2010; Walker et al., 2014; Walker et al., 2015), which may improve classification accuracy compared to using original VI values (Simonneaux and Francois, 2003; Simonneaux et al., 2008). More complex temporal feature extractors like pre-defined mathematical equations or models have been widely used for multi-temporal classification and vegetation phenology studies. In these studies, VI time series were represented by a function or a set of functions, for example, Fourier transform (Olsson and Eklundh, 1994, Verhoef et al., 1996, Evans and Geerken, 2006, M. Zhang et al., 2008, Geerken, 2009), wavelet transform (Sakamoto et al., 2005; Sakamoto et al., 2006; Galford et al., 2008), Savitzky-Golay filter (J. Chen et al., 2004, Shao et al., 2016), linear regression (Funk and Budde, 2009), spline fitting (Bradley et al., 2007; Bradley and Mustard, 2008), Hidden Markov Model (Siachalou et al., 2015), Kalman Filter (Vicente-Guijalba et al., 2014), manually-defined shapes (Sakamoto et al., 2010; Sakamoto et al., 2011; Sakamoto et al., 2013; Sakamoto et al., 2014), or a combination of function with different forms (Wang et al., 2012). Among the curve-fitting functions, the logistic/sigmoid function proposed by Badhwar (1984) has gained popularity for its robustness and convenience to derive phenological features in characterizing vegetation dynamics and growing cycles (X. Zhang et al., 2003, Fisher et al., 2006, Beck et al., 2006, Fisher and Mustard, 2007, Dannenberg et al., 2015, Xin et al., 2015, Gonsamo and Chen, 2016).

While existing approaches of temporal feature extraction provide many choices to represent vegetation dynamics, in practice it is not an easy task to find an effective and suitable approach. Some problems are:

- The manual work of model design and feature extraction relies on human experience and domain knowledge. Features from general models may not be adequate for specific tasks compared to features directly learned from data and driven by tasks in an end-to-end fashion.

- Manual feature engineering is also tedious and time-consuming. Human intervention is usually required to deal with changing environmental and weather conditions. It is hard for human knowledge to simultaneously account for the complex factors such as interclass similarity, intra-class variability, atmospheric conditions, and light-scattering mechanisms (Zhu et al., 2017).

- The fixed form of pre-defined models and associated mathematical assumptions limit the flexibility to handle disparate temporal patterns. For approaches based on curve-fitting or empirical seasonal patterns, it is difficult to choose a curve function for all the vegetation types particularly in a diverse landscape (Zhong et al., 2011).

feature extractor could be trainable to automatically adapt for specific tasks while bringing minimal mathematical constraints to handle complex and varying patterns, just as the way experienced human experts learn to interpret and identify vegetation types based on the shape of seasonal VI profiles. The human learning process to perceive and recognize temporal patterns appears quite different from aforementioned traditional methods. Human experts do not simply check values in the series by time steps like some decision-tree-based methods. They neither limit their understanding to a fixed mathematical model nor a group of models. One interesting fact is that even for experts it is almost impossible to clearly list all the rules of pattern identification, making it difficult to develop classification algorithms at the human knowledge level (Clancey, 1983; Ripley, 2007). The active field of deep learning often provides solutions to such tasks and shows great potential in feature representation of remotely-sensed time series (Zhu et al., 2017).

Inspired by the learning processing of human beings, artificial neural networks (ANNs) employ a general structure of connected units to learn feature representation exclusively from data and reduce task-specific and explicit-rule-based programming. Deep learning models, or deep ANNs with more than two hidden layers, provide sufficient model complexity to learn feature representations from data in an end-to-end regime instead of manual feature engineering based on human experience and prior knowledge (LeCun and Bengio, 1995). In recent years, deep learning was considered as a breakthrough technology in machine learning and data mining including the research field of remote sensing. Image classification studies particularly benefit from deep learning due to its flexibility in feature representation, automation by expert-free end-to-end learning, and computational efficiency. With deep learning models features are automatically extracted for given classification tasks without pre-defined feature crafting algorithms by building a part of the deep ANNs as autoencoders (Y. Chen et al., 2014, W. Li et al., 2016, Wan et al., 2017, Mou and Zhu, 2018, Mou et al., 2018a), or feature selectors (Zou et al., 2015).

Convolutional neural network (CNN) is one of the most successful network architecture in deep learning methods. The learning process of CNNs is computational efficient and insensitive to shift in data like image translation, making CNN a leading model to recognize 2D patterns in images (Krizhevsky et al., 2012). In remote sensing studies, 2D CNNs have been widely used to extract spatial features from the dimensions of width and height for object detection and semantic segmentation of high resolution images (X. Chen et al., 2014, Penatti et al., 2015, F. Hu et al., 2015, Kampffmeyer et al., 2016, Sherrah, 2016, Audebert et al., 2018, Maggiori et al., 2017, W. Li et al., 2017, Volpi and Tuia, 2017, Marcos et al., 2018, Marmanis et al., 2018). Another major application of CNNs is hyperspectral image classification, in which CNNs were used to extract spatial-spectral features, through either 1D convolution across the spectral dimension (W. Hu et al., 2015), 2D across the spatial dimensions (Yue et al., 2015; Zhao and Du, 2016; Mou et al., 2018a), or 3D across the spectral and the spatial dimensions concatenated hyperspectral images from three seasons and applied 1D convolution to the spectral domain for land cover classification. In these studies, convolutional layers in CNNs play a role as feature extractors mostly for the spatial or the spectral domains, but rarely for the temporal domain of remotely sensed time series.

Recurrent neural networks (RNN) are another category of ANNs that extend the conventional networks with loops in connections (Connor et al., 1994; Zaremba et al., 2014). RNNs are specialized for sequential data analysis and have recently shown to be successful in several remote sensing applications. Lyu et al. (2018) adopted a RNN to leverage the sequential properties of multispectral data, such as spectral correlations and band-to-band variability across the spectral dimension. Mou and Zhu (2018) used an encoder to produce multi-level convolutional feature maps from shallow to deep, and a RNN as the decoder to recursively collect multi-scale features and aggregate features sequentially into a high-resolution semantic segmentation image. RNNs are capable of representing data in continuous dimensions with sequential dependency, and the most popular use of RNNs is in the temporal domain to extract features from multi-temporal observations.

Because of their capability of analyzing sequential data, RNNs are often considered as a natural candidate to learn the temporal relationship in image time series and model land cover changing patterns (Mou et al., 2018b). Variants of RNNs are usually used for improved learning efficiency, and the most well-known variant is the long short-term memory (LSTM), a special RNN unit that represents temporal dependency at various time-spans with gated recurrent connections (Hochreiter and Schmidhuber, 1997). Lyu et al. (2016) employed a LSTM model to learn a joint spectral-temporal feature representation from a bi-temporal image pair for change detection. Mou et al. (2018b) extended the work by using 2D convolutional layers as spatial feature extractors to provide inputs to LSTM, and reported that by employing temporal dependency, the proposed network achieved better results than traditional change detection algorithms based on simple image differencing or stacking. Rußwurm and Körner (2017) used LSTM to extract dynamic temporal features from a longer image sequence to classify crop types. In their recent study, Rußwurm and Körner (2018) improved the architecture by adding 2D convolutional layers as spatial feature extractors and connecting recurrent cells in a bidirectional manner to reduce temporal biases towards later images. Jia et al. (2017) used LSTM to learn the pattern of land cover transitions and predict land cover at each time step as a sequential output. These studies also compared the classification results by LSTM-based models with other approaches to show the advantage of using LSTM as a temporal feature extractor. For example, Lyu et al. (2016) found that LSTM achieved ~95% accuracy compared to ~80% by Support Vector Machine (SVM) and ~70% by Decision Tree in multiple experiments. Mou et al. (2018b) also conducted a series of experiments, in which the combined use of CNN and LSTM resulted in ~98% accuracy that was superior to SVM (~95%) and Decision

This study aims to develop deep learning models as the temporal feature representation approach to identify crop growth patterns and classify crop types. Deep learning models were used as an integrated framework that simultaneously extracts features and performs classification following the end-to-end principle to eliminate the need for handcrafted features and pre-defined functions. Classification results of the proposed models were compared with those from some leading machine learning classifiers. The classification problem, identifying summer crop types using VI time series, is a task that human experts are relatively good at and could effectively evaluate classification results with proper visualization and mapping settings. The major contributions of this study include:

- **Exploring the feasibility of using CNNs for temporal feature representation in multi-temporal classification**. When processing time series, a RNN or its variants like LSTM is often deemed as a natural starting point as RNNs were initially proposed to analyze sequential data. In contrast, CNNs were commonly used in spatial and spectral domains but hardly across the temporal dimension in remote sensing studies, although theoretically 1D convolution is effective and computationally efficient to recognize temporal patterns at multiple scales. In this study, the input data are time series of a single vegetation index (EVI), so the information used for classification is almost purely temporal. By comparing convolutional architectures with LSTM-based models and conventional approaches, we propose that CNNs have great potential in temporal feature representation and should be considered as a candidate for multi-temporal image classification tasks.

- **Visualizing activations of deep neural networks at various temporal scales and interpreting how the deep learning model works**. One common criticism of ANNs is that they are relatively “black box”. In remote sensing studies the visualization of deep learning models is challenging and rarely done with a few exceptions. Guidici and Clark (2017) analyzed activation maps created by the convolutional layer to interpret what the classifier learned from the spectral dimension, and highlighted local spectral regions excited by the convolutional layer to explore the distinctive spectral nature of the classes. Rußwurm and Körner (2018) visualized LSTM cell activations and showed how information aggregated over the sequence. Mou and Zhu (2018) utilized class activation maps to visualize learned spatial features from local to holistic levels. In our current study, we focused on the visualization methods for temporal patterns with various lengths from ~monthly to seasonal scales and from lower to higher layers. Two state-of-art methods were attempted: one is finding the special pattern in an on how each part of the designed CNN recognizes patterns of crop seasonality and connects learned features to crop phenology and farming practices.

The paper is organized as 5 sections. Following the introduction in Section 1, Section 2 describes the methods to develop deep learning architectures for classification, optimize models, and compare various deep learning models with other classifiers, and the results are presented in Section 3. Section 4 then interprets the behavior of the optimized model to understand how it works and discusses the feasibility of using deep learning models as the representation method for time series in classification applications. Finally Section 5 concludes the paper.

Section snippets

Study area

The study area is Yolo County, California, US. Yolo County mostly locates in the Central Valley and is a major area for agricultural production. Mountainous areas on the west side of the county are unsuitable for cultivation and were excluded from further analysis using a Digital Elevation Model. The east side of the county has flat terrain and ideal soil and irrigation conditions (Fig. 1). Precipitation is about 400 mm per year and mostly occurs in winter as a characteristic of Mediterranean …

Result

The optimized MLP architecture is relatively shallow with two hidden layers, and each layer has 512 neurons. Deeper MLP models did not improve the validation set accuracy. The training time of MLP models ranged from a few minutes to 2 h on a P4000 GPU. The selected model was trained for half an hour until the validation accuracy reached a plateau. Architectures of the Conv1D-based and

Interpreting the behavior of the deep network model

The two deep neural networks we designed treat input EVI series in different ways. A LSTM cell models the time series with units changing step by step from time 1 to time *n*. A Conv1D layer has a pattern or shape template in each of its channels and matches the patterns with the input through convolution. It is valuable to look into the components of the Conv1D-based model, the top classifier of the present study, to understand how the model recognizes EVI series, which is often a difficult task …

Conclusion

In this study we developed deep neural networks with various architectures to classify summer crops using EVI time series. The most satisfactory classification result was achieved by a deep network model built with one-dimensional convolutional (Conv1D) layers and an inception module. According to the overall accuracy and the capability of identifying individual crop types, the optimal model is superior to popular classifiers such as XGBoost, Random Forest, and Support Vector Machine as well as …

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